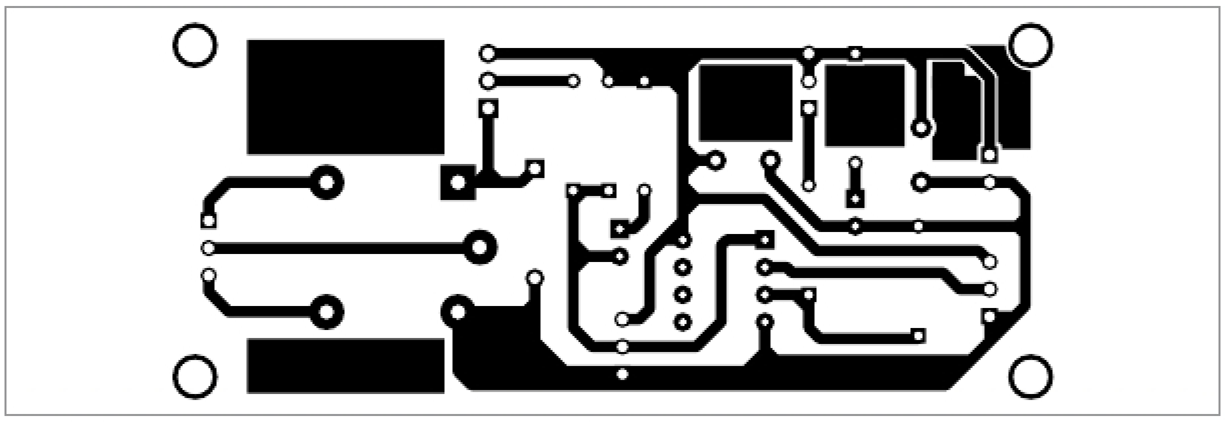

Fig. 4: Actual-size PCB layout of overcurrent fault detector

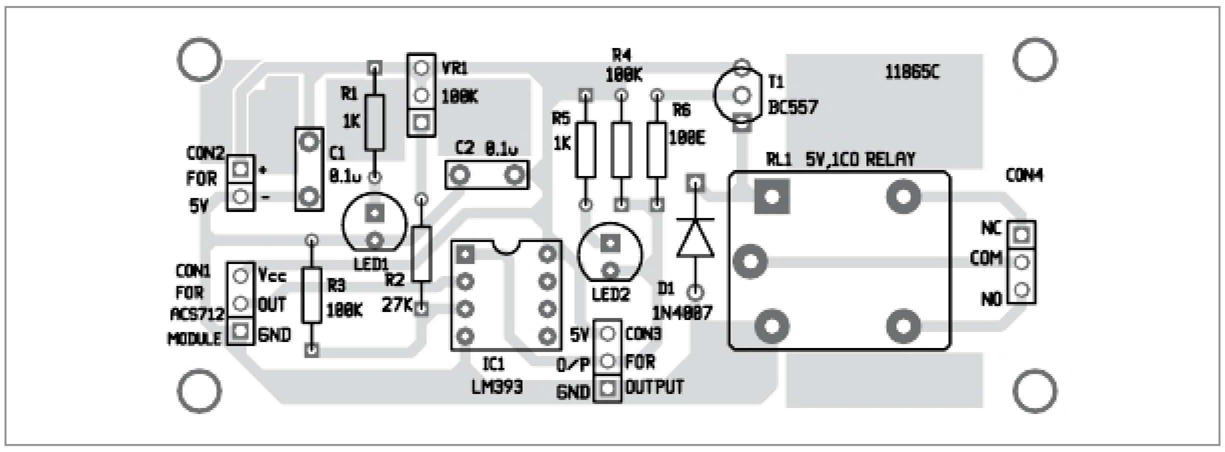

Fig. 5: Components layout of the PCB

like annunciator (or connect/disconnect any electrical device) connected across CON4 once the current reaches a particular level.

Construction and calibration

An actual-size PCB layout for the overcurrent fault detector is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit on PCB, connect a 5V across CON2. Connect a 12V battery at IP+ and load at IP- of the ACS712 sensor module.

The trip-point can easily be configured through the potmeter (VR1). The trip-point in the author's prototype is set at 3A.

To set up for a particular application, you can alter the values of VR1 and R2. Assume that you need an overcurrent fault detection just above 2A. At the current of 2A, the ACS712 module roughly gives an output of 2.87V, and the same voltage should be at its inverting input pin. Therefore you should tune VR1 to set the appropriate voltage at the non-inverting input of IC1 so that it provides an overcurrent fault alert signal (logic-low) when the current level goes beyond 2A.

Note. The calibration process demands both theoretical and empirical skills. ACS712 current sensor used here can handle current up to 5A. It can convert input current into an analogue output voltage of about 2.5V plus 185mV/A. One drawback of the ACS712 sensor is that it tends to be quite unstable at lower currents. It is hypersensitive to external magnetic fields, so keep the working circuit away from strong magnetic fields. EFY

PARTS LIST

Semiconductors:	
IC1	- LM393 operational amplifier
D1	- 1N4007 rectifier diode
T1	- BC557 pnp transistor
LED1, LED2	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R5	- 1-kilo-ohm
R2	- 27-kilo-ohm
R3, R4	- 100-kilo-ohm
R6	- 100-ohm
VR1	- 100-kilo-ohm potmeter
Capacitors:	
C1, C2	- 0.1µF ceramic disk
Miscellaneous:	
CON1, CON3, CON4	- 3 pin terminal connector
CON2	- 2 pin terminal connector
RL1	- 5V, single-changeover relay
	- ACS712 module (5A)
	- 8-pin IC base
	- 5V regulated power supply

T.K. Hareendran is an electronics designer, hardware beta tester, technical author, and product reviewer